

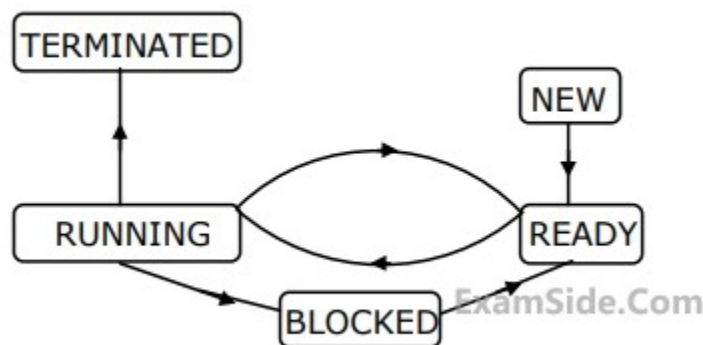
OPERATING SYSTEM GATE QUESTIONS

Process Concepts and CPU Scheduling

- 1) Which scheduling policy is most suitable for a time-shared operating systems?
- Shortest Job First
 - Round Robin
 - First come First serve
 - Elevator

Answer: Round Robin

- 2) The process state transition diagram in Figure is representative of



- A batch operating system
 - An operating system with a non-preemptive scheduler
 - An operating system with a preemptive scheduler
 - A uni-programmed operating system
- Answer: An operating system with a preemptive scheduler

- 3) Which of the following is an example of spooled device?
- A line printer used to print the output of number of jobs.
 - A terminal used to enter input data to a running program.
 - A secondary storage device in a virtual memory system
 - A graphic display device
- Answer: A line printer used to print the output of number of jobs.

- 4) Which of the following is an example of spooled device?
- The terminal used to enter the input data for the C program being executed.
 - An output device used to print the output of a number of jobs.
 - The secondary memory device in a virtual storage system.
 - The swapping area on a disk used by the swapper.
- Answer: An output device used to print the output of a number of jobs.

- 5) Consider n processes sharing the CPU in a round-robin fashion. Assuming that each process switch takes s seconds, what must be the quantum size q such that the overhead resulting from process switching is minimized but, at the same time each process is guaranteed to get its turn at the CPU at least every t seconds?

A $q \leq \frac{t - ns}{n - 1}$

B $q \geq \frac{t - ns}{n - 1}$

C $q \leq \frac{t - ns}{n + 1}$

D $q \geq \frac{t - ns}{n + 1}$

Answer:

A $q \leq \frac{t - ns}{n - 1}$

- 6) System calls are usually invoked by using:

- a) A software interrupt
- b) Polling
- c) An indirect jump
- d) A privileged instruction

Answer: A software interrupt

- 7) A processor needs software interrupt to

- a) Test the interrupt system of the processor
- b) Implement co-routines
- c) Obtain system services which need execution of privileged instructions
- d) Return from subroutine

Answer: Obtain system services which need execution of privileged instructions.

- 8) Consider a set of n tasks with known runtimes $r_1, r_2, \dots, r_n$ to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?

- a) Round-Robin
- b) Shortest -Job- First
- c) Highest-Response-Ratio-Next
- d) First-Come-First-Served

Answer: Shortest -Job- First

- 9) The maximum number of processes that can be in Ready state for a computer system with n CPUs is :

- a) n
- b) n^2
- c) $2n$
- d) Independent of n

Answer: d

10) Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

- (A) Shortest remaining time first
- (B) Round-robin with time quantum less than the shortest CPU burst
- (C) Uniform random
- (D) Highest priority first with priority proportional to CPU burst length

Answer: (A)

11) The following C program is executed on a Unix / Linux system:

```
#include <unistd.h>
```

```
int main() {
    int i;
    for (i = 0; i < 10; i++)
        if (i % 2 == 0) fork();
    return 0;
}
```

The total number of child process created is _____ .

Note – This was Numerical Type question.

- (A) 31
- (B) 63
- (C) 5
- (D) 6

Answer: (A)

Synchronization and Concurrency

1 mark:

1. critical section is a program segment
 - (A) which should run in a certain specified amount of time
 - (B) which avoids deadlocks
 - (C) where shared resources are accessed
 - (D) which must be enclosed by a pair of semaphore operations, P and V

Answer: (C)

2. What is said to happen when the result of computation depends on the speed of the processes involved?

- a. A deadlock
- b. A time lock
- c. Cycle stealing
- d. Race condition

Answer: Race condition

3. Let $m[0] \dots m[4]$ be mutexes (binary semaphores) and $P[0] \dots P[4]$ be processes. Suppose each process $P[i]$ executes the following:

```
wait (m[i]); wait(m[(i+1) mode 4]);
```

```
-----
```

```
release (m[i]); release (m[(i+1)mod 4]);
```

This could cause:

- (A) Thrashing
- (B) Deadlock
- (C) Starvation, but not deadlock
- (D) None of the above

Answer: (B)

4. A counting semaphore was initialized to 10. Then 6P (wait) operations and 4V (signal) operations were completed on this semaphore. The resulting value of the semaphore is

- 0
- 8
- 10
- 12

Answer: (B)

5. Where does the swap space reside?

- (a) RAM
- (b) Disk

- (c) ROM
 - (d) On-chip cache
- Answer: (b)

6. Suppose a processor does not have any stack pointer register. Which of the following statements is true?

- (A) It cannot have subroutine call instruction
- (B) It can have subroutine call instruction, but no nested subroutine calls
- (C) Nested subroutine calls are possible, but interrupts are not
- (D) All sequences of subroutine calls and also interrupts are possible

Answer: (A)

7. Which of the following scheduling algorithms is non-preemptive?

- (A) Round Robin
- (B) First-In First-Out
- (C) Multilevel Queue Scheduling
- (D) Multilevel Queue Scheduling with Feedback

Answer: (B)

8. Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned.

Method Used by P1

while (S1 == S2) ;

Critical Section

S1 = S2;

Method Used by P2

while (S1 != S2) ;

Critical Section

S2 = not (S1);

Which one of the following statements describes the properties achieved?

- (A) Mutual exclusion but not progress
- (B) Progress but not mutual exclusion
- (C) Neither mutual exclusion nor progress
- (D) Both mutual exclusion and progress

Answer: (A)

9. Three concurrent processes X, Y, and Z execute three different code segments that access and update certain shared variables. Process X executes the P operation (i.e., wait) on semaphores a, b and c; process Y executes the P operation on semaphores b, c and d; process Z executes the P operation on semaphores c, d, and a before entering the respective code segments. After completing the execution of its code segment, each process invokes the V operation (i.e., signal) on its three semaphores. All

semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the P operations by the processes?

- (A) X: P(a)P(b)P(c) Y: P(b)P(c)P(d) Z: P(c)P(d)P(a)
- (B) X: P(b)P(a)P(c) Y: P(b)P(c)P(d) Z: P(a)P(c)P(d)
- (C) X: P(b)P(a)P(c) Y: P(c)P(b)P(d) Z: P(a)P(c)P(d)
- (D) X: P(a)P(b)P(c) Y: P(c)P(b)P(d) Z: P(c)P(d)P(a)

Answer: (B)

10. A file is organized so that the ordering of data records is the same as or close to the ordering of data entries in some index. Then that index is called

- (A) Dense
- (B) Sparse
- (C) Clustered
- (D) Unclustered

Answer: (C)

11. The following two functions P1 and P2 that share a variable B with an initial value of 2 execute concurrently.

```
P1()
{
    C = B - 1;
    B = 2*C;
}
```

```
P2()
{
    D = 2 * B;
    B = D - 1;
}
```

The number of distinct values that B can possibly take after the execution is

- (A) 3
- (B) 2
- (C) 5
- (D) 4

Answer: (A)

Deadlocks

1. A computer has six tape drives, with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free? Ans: 5
2. A system has 6 identical resources and N processes competing for them. Each process can request at most 2 resources. Which one of the following values of N could lead to a deadlock? Ans: 6
3. Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is _____. Ans: 2

Memory Management

1. Which page replacement policy sometimes leads to more page faults when size of memory is increased? Ans: FIFO
2. In a paged segmented scheme of memory management, the segment table itself must have a page table because: Ans: The segment table is too large to fit in one page
3. The principle of locality justifies the use of Ans: Caches
4. A linker is given object modules for a set of programs that were compiled separately. What information needs to be included in an object module? Ans: Absolute address of internal symbols
5. A ROM is used to store the table for multiplication of two 88-bit unsigned integers. The size of ROM required is Ans: $64K \times 16$
6. Locality of reference implies that the page reference being made by a process Ans: Is likely to be to one of the pages used in the last few page references
7. Thrashing
Ans: Implies excessive page I/O
8. Dirty bit for a page in a page table
Ans: Helps avoid unnecessary writes on a paging device
9. In a resident-OS computer, which of the following systems must reside in the main memory under all situations? Ans: Loader
10. Which of the following statements is false?
Ans: Virtual memory implements the translation of a program's address space into physical memory address space.
11. The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called
Ans: Relocation
12. Consider a virtual memory system with FIFO page replacement policy. For an arbitrary page access pattern, increasing the number of page frames in main memory will
Ans: sometimes increase the number of page faults
13. Which of the following is not a form of memory?
Ans: Instruction opcode

14. The optimal page replacement algorithm will select the page that.

Ans: Will not be used for the longest time in the future.

15. In a system with 32 bit virtual addresses and 11 KB page size, use of one-level page tables for virtual to physical address translation is not practical because of

Ans: The large memory overhead in maintaining page tables.

16. Consider a program P that consists of two source modules M1 and M2 contained in two different files. If M1 contains a reference to a function defined in M2, the reference will be resolved at.

Ans: Link-time

17. Which of the following addressing modes are suitable for program relocation at run time?

1. Absolute addressing
2. Based addressing
3. Relative addressing
4. Indirect addressing

Ans: 2 and 3

18. The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by.

Ans: the instruction set architecture

19. What is the swap space in the disk used for?

Ans: Saving process data

20. In which one of the following page replacement policies, Belady's anomaly may occur?

Ans: FIFO

21. The essential content(s) in each entry of a page table is / are:

Ans: Page frame number.

22. How many 32K×1RAM chips are needed to provide a memory capacity of 256 K-bytes?

Ans: 64

23. A system uses FIFO policy for page replacement. It has 4 pages frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?

Ans: 196

24. A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page

frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below? 4,7,6,1,7,6,1,2,7,2

Ans: 6

25. Consider a system with byte-addressable memory, 32-bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____.

Ans: 4

26. A computer system implements a 40-bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLBtag in bits is _____.

Ans: 22

27. In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

Ans: FIFO

28. Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

Ans: The hole created by best fit is never larger than the hole created by first fit.

File System IO and Protection

1. I/O redirection

Ans: Can be employed to use an existing file as input file for a program

2. The correct matching for the following pairs is

List - II

(a) DMADMA I/OI/O

(b) Cache

(c) Interrupt I/OI/O

(d) Condition Code Register

List - III

(1) High speed RAMRAM

(2) Disk

(3) Printer

(4) ALU

Ans: a-2, b-1, c-3, d-4

3. When an interrupt occurs, an Operating System

Ans: Always resumes execution of interrupted process after processing the interrupt

4. The correct matching for the following pairs is

(a) Disk scheduling (1) Round robin

(b) Batch processing (2) SCAN
(c) Time sharing (3) LIFO
(d) Interrupt processing (4) FIFO
Ans: a-2,b-4,c-1,d-3

5. Which of the following devices should get higher priority in assigning interrupts?
Ans: keyboard

6. Which of the following is true?
Ans: A processor checks for interrupts before executing a new instruction.

7. Listed below are some operating system abstractions (in the left column) and the hardware components. Which matching pairs is correct?

List-II

(a) Thread (b) Virtual Address space
(c) File System (d) Signal

List-III

(1) Interrupt (2) Memory
(3) CPU (4) Disk

Ans: a-3,b-2,c-4,d-1

8. Which of the following disk scheduling strategies is likely to give the best throughput?
Ans: Nearest cylinder next

9. Which of the following requires a device driver?
Ans: disk

10. Using a larger block size in a fixed block size file system leads to
Ans: better disk throughput but poorer disk space utilization

11. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs?
Ans: 0%

12. Normally user programs are prevented from handling I/O directly by I/O instructions in the for CPU having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privilege in CPU with memory mapped I/O there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O?
Ans: I/O protection ensured by operating system routines(s)

13. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively: A

Ans: 256 Mbytes, 19 bits

14. The data blocks of a very large file in the Unix file system are allocated using

Ans: an extension of indexed allocation.

15. Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests.

Correct Answer is 3

16. A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____.

Correct Answer is 99.6